KCPC Workshop 1
Solutions

1 Introductory brainteaser

1.1 Left handed people

Answer: 50 people must leave.

Explanation: Initially, there are 99 left-handed people and 1 right-handed person. To make the 1 right-handed person represent 2% of the total population, we solve for the new total population x:

$$0.02x = 1 \implies x = 50$$

Since the new total must be 50, then $100 - 50 = 50$ people must leave.

2 Induction and Logic Puzzles

2.1 The Locker Problem

> **Key Insight:** A locker is toggled once for every divisor it has. Only perfect squares have an odd number of divisors.

Solution: A locker n ends up open if it is toggled an odd number of times. This only happens if n is a perfect square.

- **Q1:** Between 1 and 100, the perfect squares are $1^2, 2^2, \ldots, 10^2$. Thus, 10 lockers are open.
- **Q2:** For N lockers, the number of open lockers is $\lfloor \sqrt{N} \rfloor$.

2.2 The Three Wise Men

Solution: The man announced he was wearing a **Blue** hat.

Proof. If there was only 1 blue hat, the wearer would see two white hats and instantly know he had blue (since at least one is blue).

If there were 2 blue hats, one wearer would see one blue and one white. He would wait to see if the other blue-hat-wearer jumped up. When they didn't (meaning they saw more than one blue), he would realize he must be the second blue hat.

Since they sat for a "very long time," the winner realized that even the 2-hat logic had failed, meaning there must be 3 blue hats. □

2.3 Muddy Children

Solution: They step forward on the m-th turn, where m is the number of muddy children.

Assuming that each child has-and knows each of the others to have-perfect logic all children with muddy faces (X) will step forward together on turn X. The children have different information, depending on whether their own face is muddy or not. Each member of X sees $X - 1$ muddy faces, and knows that those children will step forward on turn $X - 1$ if they are the only muddy faces. When that does not occur, each member of X knows that he or she is also a member of X, and steps forward on turn X. Each non-member of X sees X muddy faces, and will not expect anyone to step forward until at least turn X.

3 Invariance and Physics

3.1 Ants on a Ruler

Key Insight: When two ants of the same speed collide and bounce, it is mathematically identical to the ants passing through each other.

Solution: Since the ants are indistinguishable and travel at constant speed, we can treat collisions as "passing through." The worst-case scenario is an ant starting at one end and traveling the full length L.

$$T = \frac{L}{v}$$

3.2 The Fly and the Cyclists

Solution: The trick is to ignore the fly's complex path and focus on **time**. The cyclists are 20 miles apart closing at a combined speed of $10 + 10 = 20$ mph. They will collide in exactly 1 hour. Since the fly travels at 15 mph for that entire hour, the distance is:

$$d = 15 \text{ mph} \times 1 \text{ hour} = 15 \text{ miles}$$

4 Game Theory

4.1 Simple Stick Game

Solution: A **winning state** is a state where the player will win the game if they play optimally, and a **losing state** is a state where the player will lose the game if the opponent plays optimally. It turns out that we can classify all states of a game so that each state is either a winning state or a losing state.

In the above game, state 0 is clearly a losing state, because the player cannot make any moves. States 1, 2 and 3 are winning states, because we can remove 1, 2 or 3 sticks and win the game. State 4, in turn, is a losing state, because any move leads to a state that is a winning state for the opponent.

More generally, if there is a move that leads from the current state to a losing state, the current state is a winning state, and otherwise the current state is a losing state. Using this observation, we can classify all states of a game starting with losing states where there are no possible moves.

It is easy to analyze this game: a state k is a losing state if k is divisible by 4, and otherwise it is a winning state. An optimal way to play the game is to always choose a move after which the number of sticks in the heap is divisible by 4. Finally, there are no sticks left and the opponent has lost.